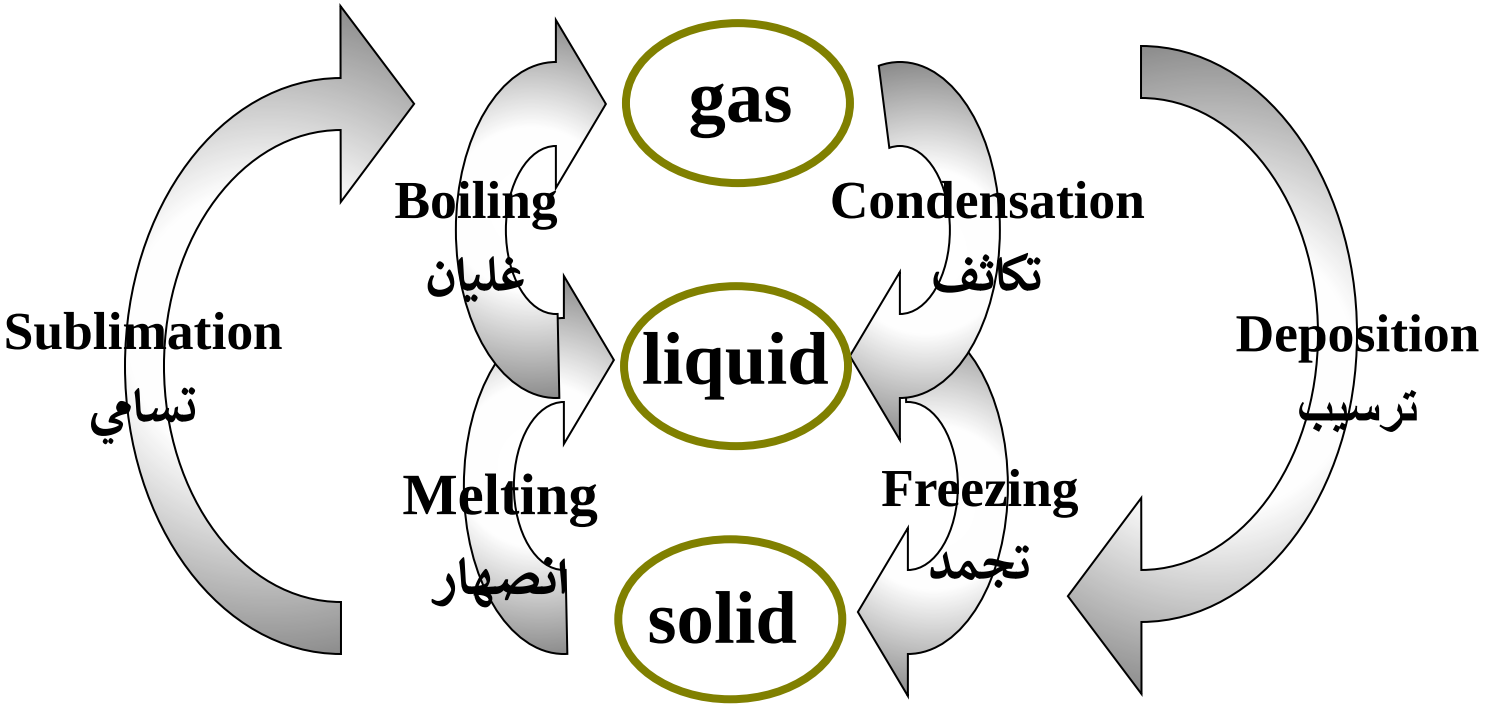
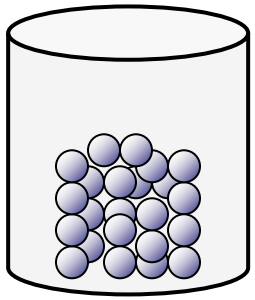
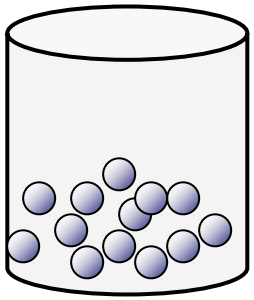
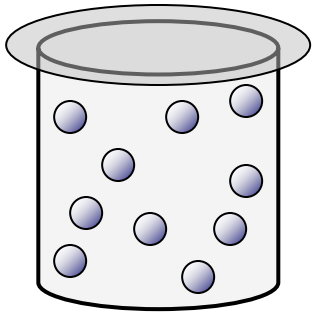
State of matter

الدكتور
عبدالله إبراهيم مفتاح
قسم الكيمياء- كلية العلوم- جامعة المنيا

States of Matter



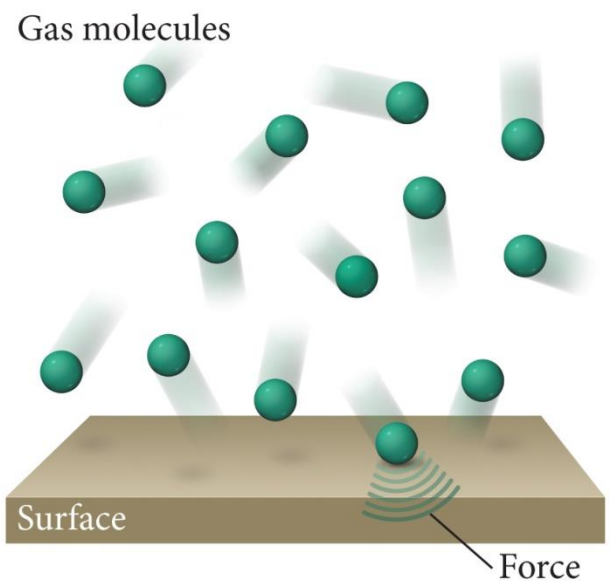
Some Properties of Solids, Liquids, and Gases

Property	Solid	Liquid	Gas
			
Shape	Has definite shape	Takes the shape of the container	Takes the shape of its container
Volume	Has a definite volume	Has a definite volume	Fills the volume of the container
Arrangement of Particles	Fixed, very close	Random, close	Random, far apart
Interactions between particles	Very strong	Strong	Essentially none

Gas Pressure

Pressure is the force exerted per unit area by gas molecules as they strike the surfaces around them.

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$



Collisions with surfaces create pressure.

- Gas pressure is a result of the constant movement of the gas molecules and their collisions with the surfaces around them.

- The pressure of a gas depends on several factors:
  - ✓ Number of gas particles in a given volume
  - ✓ Volume of the container
  - ✓ Average speed of the gas particles

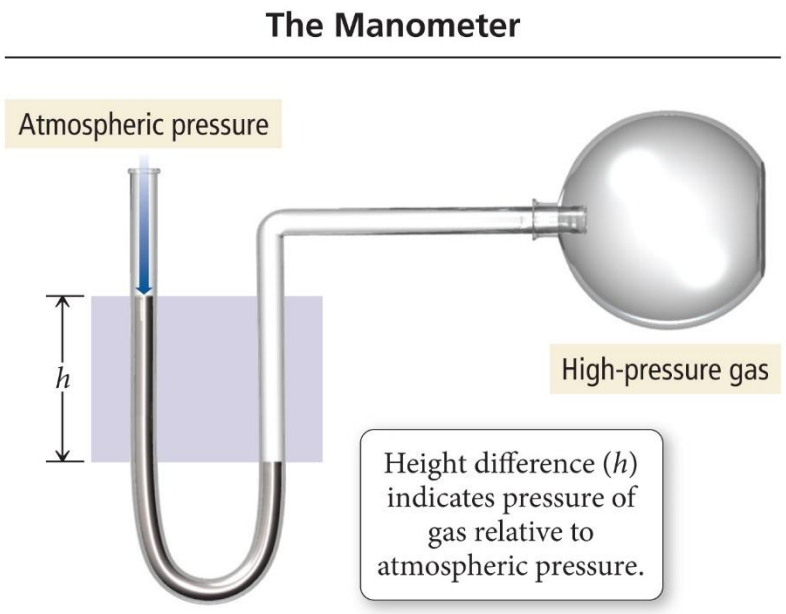
The Manometer

- The pressure of a gas trapped in a container can be measured with an instrument called a **manometer.**

- Manometers are U-shaped tubes partially filled with a liquid that are connected to the gas sample on one side and open to the air on the other.

- A competition is established between the pressures of the atmosphere and the gas.

- The difference in the liquid levels is a measure of the difference in pressure between the gas and the atmosphere.



For this sample the gas pressure is *greater than* atmospheric pressure, the mercury level on the left side of the tube is *higher than* the level on the right.

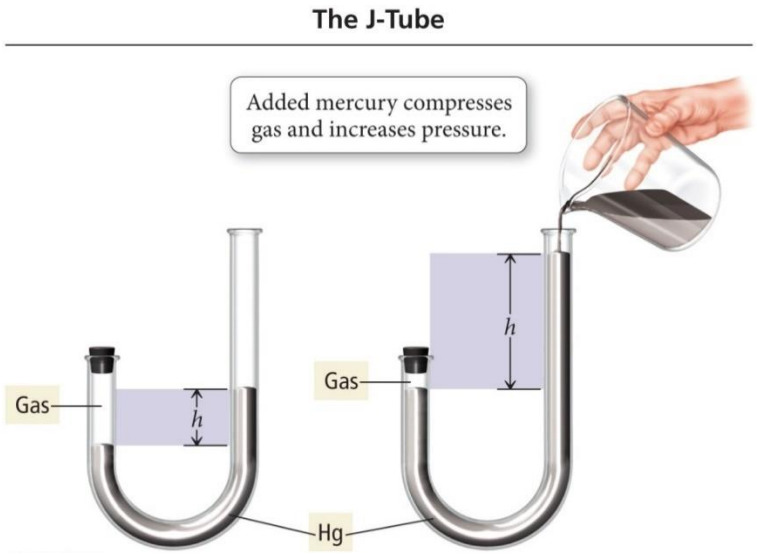
Boyle's Law: Robert Boyle (1627–1691)

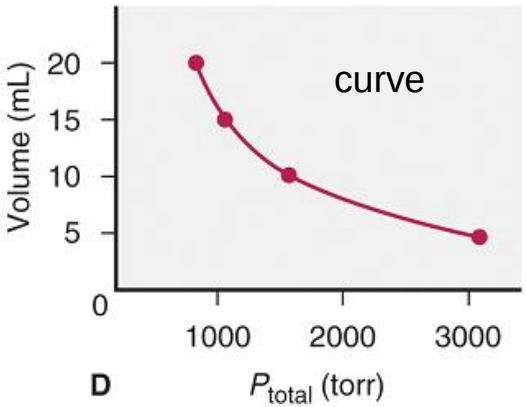
- Pressure of a gas is inversely proportional to its volume (Constant T and amount of gas).

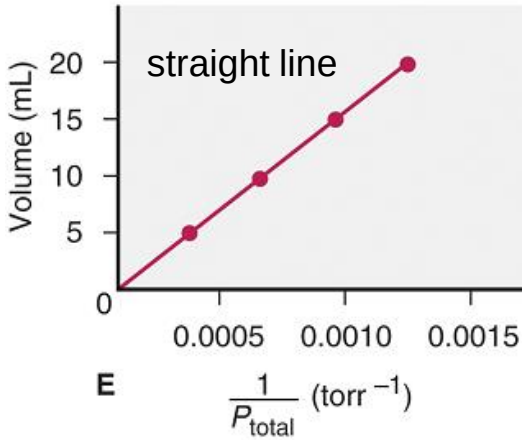
$$P \times V = \text{constant}$$

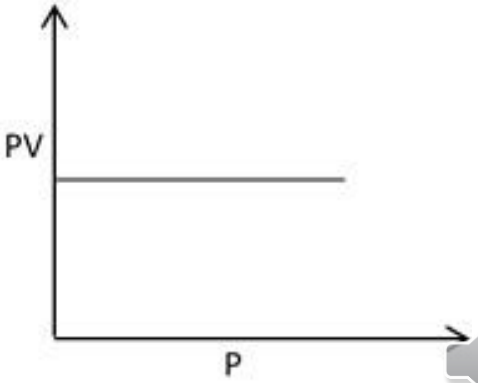
$$P_1 \times V_1 = P_2 \times V_2$$

- Robert Boyle and Robert Hooke used a J-tube to measure the volume of a sample of gas at different pressures.
- They trapped a sample of air in the J-tube and added mercury to increase the pressure on the gas.
- They observed an inverse relationship between volume and pressure.





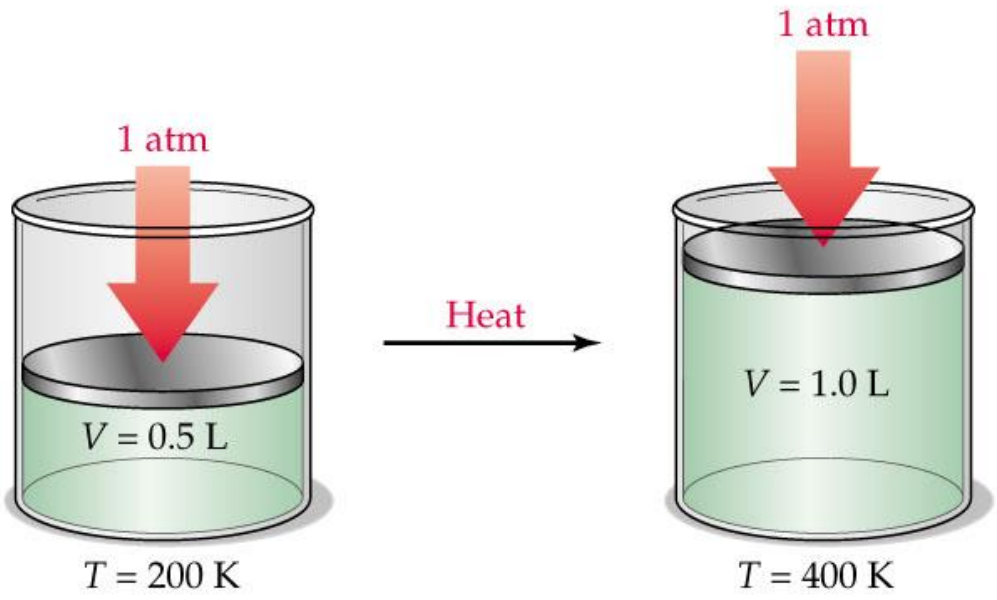




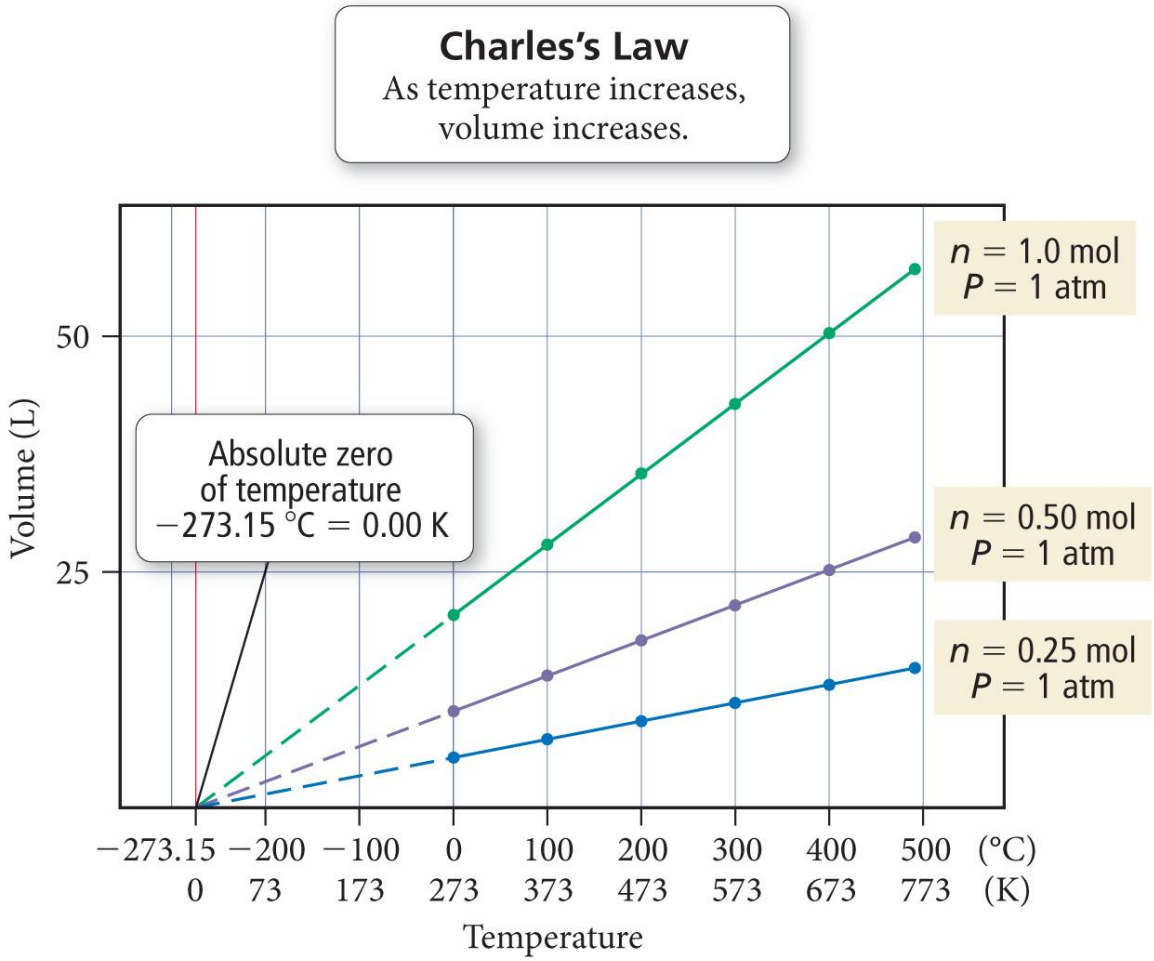
Charles's Law: Volume and Temperature

- The volume of a fixed amount of gas at a constant pressure increases linearly with increasing temperature in kelvins:
  - ✓ The volume of a gas increases with increasing temperature.
- Kelvin T = Celsius T + 273
- V = constant × T

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$



Charles's Law



If the lines are extrapolated back to a volume of "0," they all show the same temperature, −273.15 °C = 0 K, called **absolute zero**

The extrapolated lines cannot be measured experimentally because all gases condense into liquids before –273.15 °C is reached.

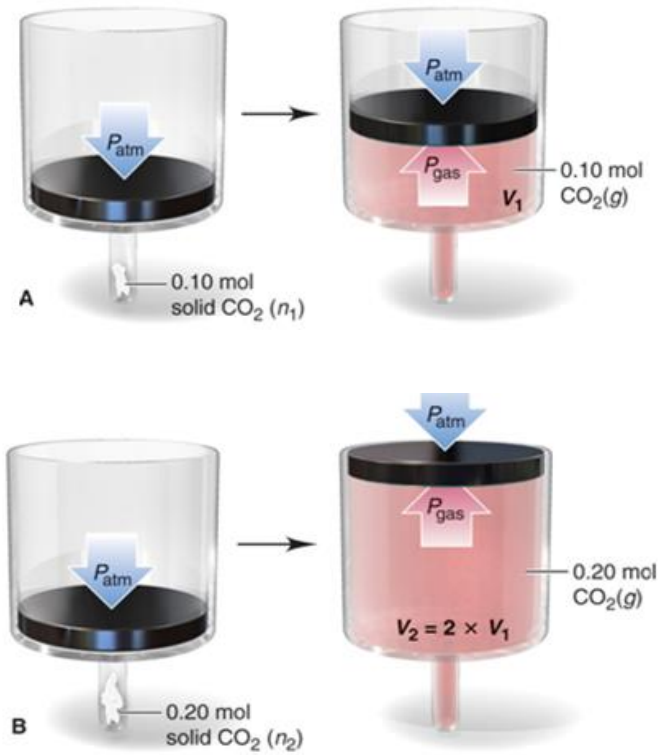
Avogadro's Law, Amedeo Avogadro (1776–1856)

- Volume directly proportional to the number of gas molecules

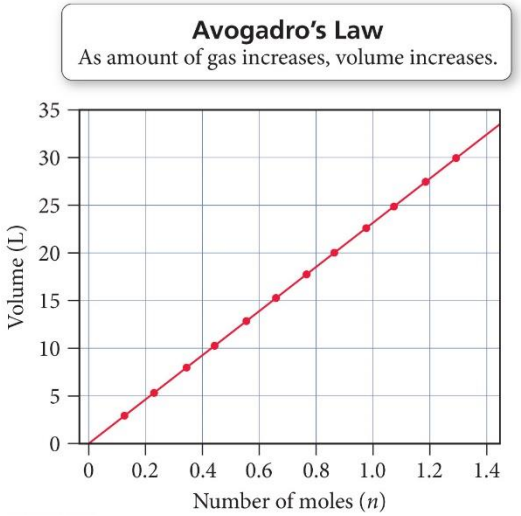
V = constant × n

Constant P and T

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$



- Equal volumes of different gases contain equal numbers of molecules (moles) (***The gas doesn't matter).***



When the amount of gas in a sample increases at constant temperature and pressure, its volume increases in direct proportion because the greater number of gas particles fill more space.

The volume of a gas sample increases linearly with the number of moles of gas in the sample.